



PERCENTAGE

GIRRAWEEEN HIGH SCHOOL
Mathematics Extension 1

HSC ASSESSMENT
HALF YEARLY
ANSWERS COVER SHEET

Name: _____

QUESTION	MARK	E2	E3	E4	E5	E6	E7
1	/20						✓
Total	/20						
2(a) - (c)	/17						✓
(d)	/13					✓	✓
Total	/30						
3(a)	/3					✓	✓
(b) (c)	/12	✓					✓
(d)	/6		✓				✓
Total	/21						
4(a)	/5	✓					✓
(b)(c)(d)i	/10						✓
(d) ii	/3					✓	✓
(e)	/6		✓				✓
	/24						
5(a) -(d)i	/14						✓
(d) (ii)	/3					✓	✓
(e)	/8						✓
	/25						
TOTAL	/120	/13	/12	/0	/0	/24	/120



GIRRAWEE HIGH SCHOOL

EXAMINATION

2008

MATHEMATICS

EXTENSION 1

*Time allowed - Two hours
(Plus 5 minutes' reading time)*

HALF YEARLY

DIRECTIONS TO CANDIDATES

- Attempt ALL questions.
- All necessary working should be shown in every question.
Marks may be deducted for careless or badly arranged work.
- Standard integrals are on the sheets supplied
- Board-approved calculators may be used.
- Each question attempted is to be returned on a *separate* piece of paper clearly marked Question 1 , Question 2 , etc. Each piece of paper must show your name.
- You may ask for extra pieces of paper if you need them.



Total marks – 118

Attempt Questions 1 – 5

All questions are NOT of equal value

Answer each question on a separate piece of paper clearly marked Question 1, Question 2 etc.

Each piece of paper must show your name.

Question 1 (20 Marks)	Use a separate piece of paper	Marks
(a) Solve for x $2^x = 9$		2
(b) Solve for x , $\frac{2x - 5}{3x + 1} \geq 1$		2
(c) Find the acute angle between the lines $2x + 3y - 7 = 0$ and $3x - y + 6 = 0$		3
(d) Find $\lim_{x \rightarrow 0} \frac{\sin 5x}{\sin 3x}$		2
(e) Find the coordinates of the point that divides the interval A(2, - 3) and B(-4 , 7) internally in the ratio 3:2.		3
(f) Sketch the graph of $y = \sin x$ in the domain $0 \leq x \leq 2\pi$. On the same set of axes and in the same domain sketch $y = \operatorname{cosec} x$		4
(g) Given that $\cos 2\theta = \cos^2 \theta - \sin^2 \theta$		
(i) Find an expression for $\cos 2\theta$ in terms of $\cos^2 \theta$		1
(ii) Hence or otherwise find the exact value of $\cos 72^\circ$ given that $\cos 36^\circ = \frac{\sqrt{5} + 1}{4}$		3

Question 2 (30 marks) Use a separate piece of paper

(a) Differentiate the following

(i) $y = xe^x$

(ii) $y = \frac{e^x}{\tan x}$ 4

(iii) $y = \ln(\sin x)$

(iv) $y = \ln \left[\frac{(x^2 - 7x)}{\sqrt{x}} \right]$ 4

(b) (i) Differentiate $y = \sin^3 x$ 2

(ii) Hence or otherwise find $\int 3 \cos x \sin^2 x dx$ 2

(c) Find the equation of the tangent and the normal to the curve $y = \tan x$, at the point

where $x = \frac{\pi}{4}$ 5

(d) Find

(i) $\int \frac{x^2 - 2x}{x^3 - 3x^2 + 7} dx$ 2

(ii) $\int_0^{\frac{\pi}{8}} \sec^2 2x dx$

(iii) $\int_1^5 \sqrt{2x-1} dx$ 6

(iv) $\int_2^3 \frac{x^2}{x^3 - 2} dx$

(v) $\int_0^{\pi} \cos x dx$ 5

Question 3 (19 Marks) Use a separate piece of paper

Marks

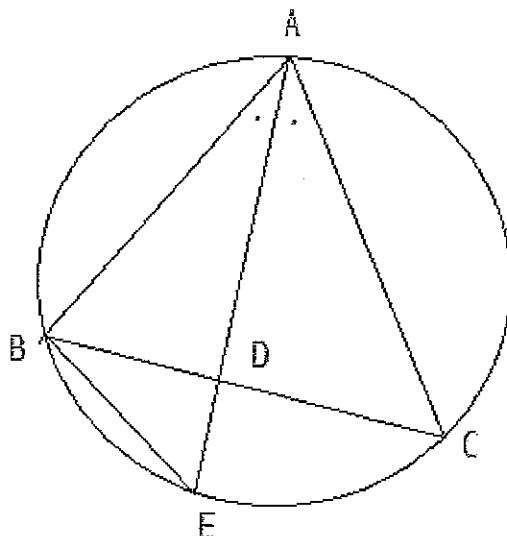
(a) Find the volume when $y = \tan x$ is rotated about the x -axis from $x = 0$ to $x = \frac{\pi}{4}$ 3

(b) Prove by Mathematical Induction that

$$1 + 8 + 27 + 64 + \dots + n^3 = \frac{1}{4} \{n^2(n+1)^2\}$$

is true for all positive values of the integer n 4

(c) In the diagram below the bisector AD of $\angle BAC$ has been extended to intersect the circle ABC at E .



(i) Copy the diagram into your solutions and prove that the $\triangle ABE$ and $\triangle ADC$ are similar. 2

(ii) Show that $AB \times AC = AD \times AE$ 2

(iii) Show that $\triangle ABD$ is similar to $\triangle CDE$ 2

(iv) Prove that $AD^2 = AB \times AC - BD \times DC$ 2

(d) In a certain country the probability of a child having blue eyes is $\frac{1}{3}$.

(i) If a family has four children what is the probability that at least 2 children will have blue eyes. 3

(ii) If 4 such families are surveyed, using your result from part (i) or otherwise, what is the probability that exactly 1 family will have at least 2 children with blue eyes. 3

Question 4 (24 Marks) Use a separate piece of paper

Marks

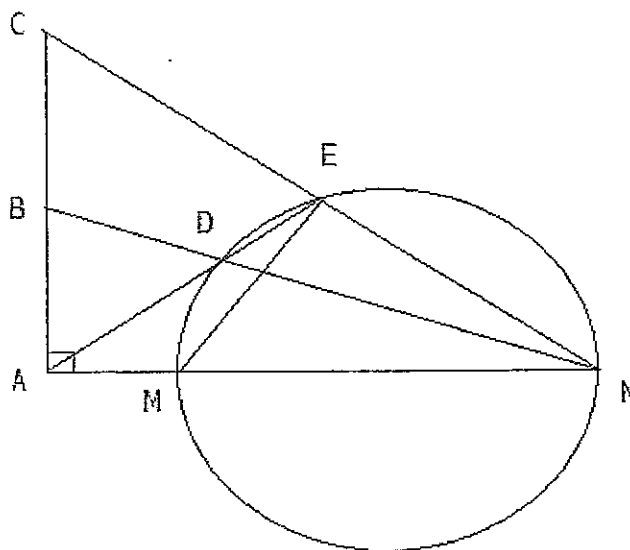
(a) In the figure below M, N, E and D are points on a circle with MN as diameter. NE is produced to C and NM is produced to A , such that CA and AN are perpendicular. AE meets the circle at D and ND is produced to meet CA at B .

(i) Show that $ACEN$ are concyclic.

2

(ii) Show that $BCED$ are concyclic.

3



(b) Find the term independent of x in the expansion of the following expression

Leave your answer in unexpanded form.

4

$$\left(2x^2 - \frac{3}{x}\right)^9$$

(c) Use the t method, where $t = \tan \frac{\theta}{2}$, to solve the following. Give your answer in radians to 2 decimal places $0 \leq \theta \leq 2\pi$

$$5 \sin \theta - 2 \cos \theta = 3$$

4

(d) (i) Differentiate the function $y = 2^x$

2

(ii) Hence or otherwise evaluate $\int_1^3 2^x dx$

3

(e) (i) How many different 9 letter "words" can be formed by using all 9 letters of the word GIRRAWEEEN

2

(ii) How many different 5 letter selections and how many different arrangements of the 5 letters can be made.

4

Question 5 (25 Marks) Use a separate piece of paper

Marks

- (a) Using the formula $f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$ differentiate $f(x) = \frac{1}{x}$,
from first principles clearly showing all necessary steps. 3
- (b) (i) Show that the first derivative of $f(x) = \sec x$ is $\frac{dy}{dx} = \sec x \tan x$ 2
(ii) Hence or otherwise find the second derivative of $f(x) = \sec x$ 2
(ii) For what values of θ , $0 \leq \theta \leq 2\pi$ is $\sec x$ concave up 2
- (c) Solve for θ , $0 \leq \theta \leq 2\pi$ the equation $3 \tan^5 \theta - 10 \tan^3 \theta + 3 \tan \theta = 0$ 4
- (d) (i) Show that $\frac{1}{x-2} - \frac{1}{x+3} = \frac{5}{(x-2)(x+3)}$ 1
(ii) Using part (i) or otherwise show that the area under $y = \frac{1}{(x-2)(x+3)}$
between $x = 3$ and $x = 5$ is $\frac{2}{5} \ln \frac{3}{2} u^2$ 3
- (e) (i) Show that $2 \sin \theta + 2\sqrt{3} \cos \theta$ can be expressed in the form $R[\sin(\theta + \alpha)]$
clearly showing the values of R and α 3
(ii) Sketch your graph of $y = R[\sin(\theta + \alpha)]$ in the range $0 \leq \theta \leq 2\pi$ 3
(iii) Solve graphically the equation $R[\sin(\theta + \alpha)] = 3$ 2



YEAR 12 HALF YEARLY 2008

SOLUTIONS

Q1 (a) $2^x = 9$

$$x \ln 2 = \ln 9$$

$$x = \frac{\ln 9}{\ln 2}$$

$$x = 3.17 \quad (2)$$

(b) $\frac{2x-5}{3x+1} \geq 1$

$$(2x-5)(3x+1) \geq (3x+1)^2$$

$$0 \geq (3x+1)^2 - (3x+1)(2x-5)$$

$$0 \geq (3x+1) \{ 3x+1 - 2x+5 \}$$

$$0 \geq (3x+1) \{ x+6 \}$$

$$-6 \leq x \leq -\frac{1}{3} \quad (2)$$

(c) $2x+3y-7=0$

$$3y = -2x+7$$

$$y = -\frac{2}{3}x + \frac{7}{3} \quad m_1 = -\frac{2}{3}$$

$$3x-y+6=0$$

$$y = 3x+6 \quad m_2 = 3$$

$$\tan \theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right|$$

$$= \left| \frac{3 + \frac{2}{3}}{1 + 3(-\frac{2}{3})} \right|$$

$$\tan \theta = \frac{11}{3}$$

$$\theta = 74.74^\circ \quad (3)$$

(d) $\lim_{x \rightarrow 0} \frac{\sin 5x}{\sin 3x}$

$$= \lim_{x \rightarrow 0} \frac{\sin 5x}{5x} \cdot \frac{3x}{\sin 3x} \cdot \frac{5x}{3x}$$

$$= 1 \cdot 1 \cdot \frac{5}{3} = \frac{5}{3} \quad (2)$$

(e) $(2, -3) \quad (-4, 7)$

$$3:2$$

$$\left(\frac{2 \times 2 + 3 \times -4}{3+2}, \frac{2 \times -3 + 3 \times 7}{2+3} \right)$$

$$= \left(-\frac{8}{5}, 3 \right) \quad (3)$$

(f) See grid paper

(g)(i) $\cos 2\theta = \cos^2 \theta - (1 - \cos^2 \theta)$

$$= 2\cos^2 \theta - 1 \quad (1)$$

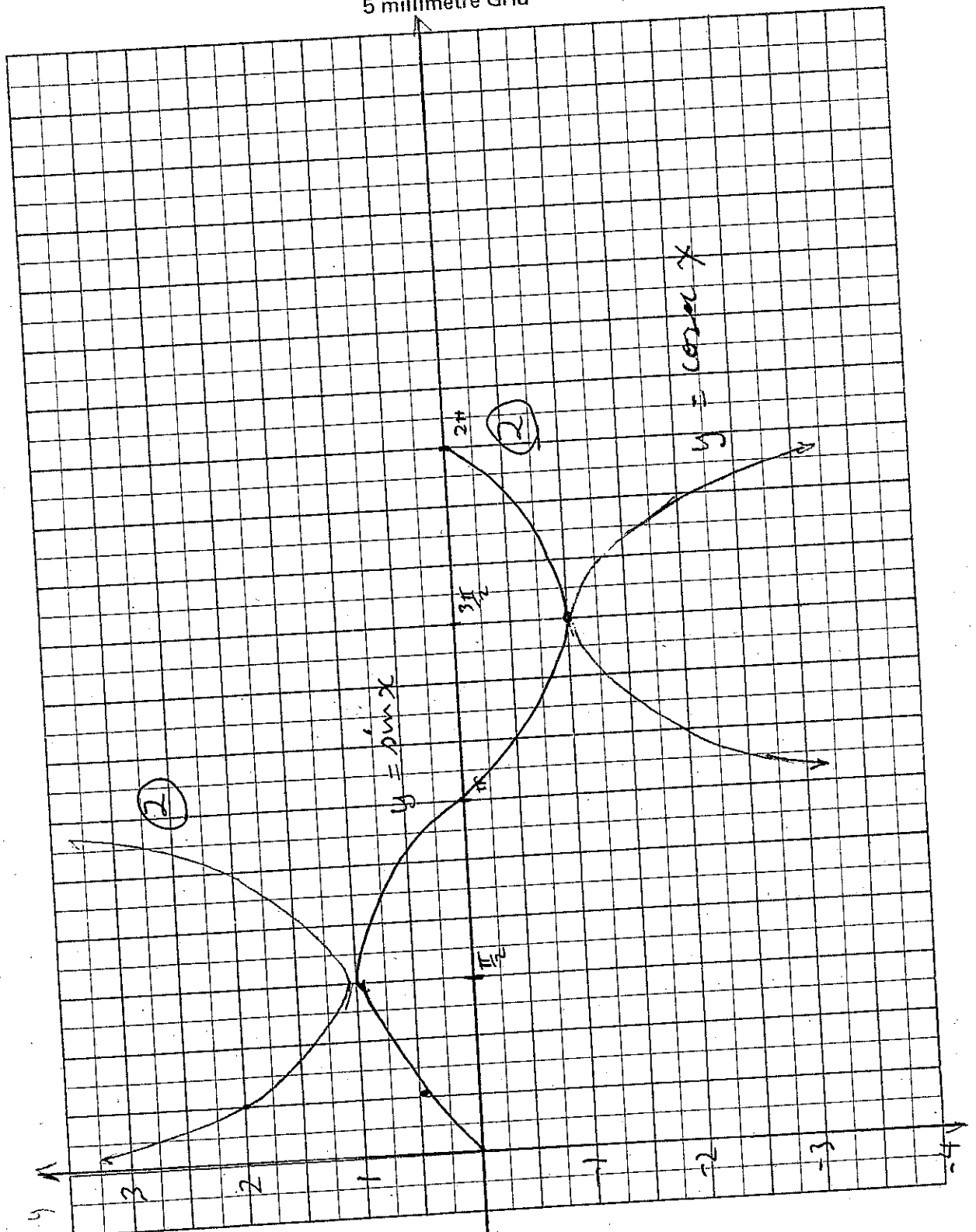
(ii) $\cos 72^\circ = 2\cos^2 36^\circ - 1$

$$= 2\left(\frac{\sqrt{5}+1}{4}\right)^2 - 1$$

$$= 2\left(\frac{6+2\sqrt{5}}{16}\right) - 1$$

$$= \frac{3+\sqrt{5}}{4} - \frac{4}{4} = \frac{\sqrt{5}-1}{4} \quad (3)$$

5 millimetre Grid



2008 EXT 1 1/2 yr 2008

Q2.(a) (i) $y = xe^x$

$$\frac{dy}{dx} = e^x + xe^x \quad (2)$$

$$\frac{dy}{dx} \Big|_{(x=\pi/4)} = 2 \quad (1)$$

Eqⁿ TANGENT

(ii) $y = \frac{e^x}{\tan x}$

$$y-1 = 2(x-\pi/4)$$

$$y = 2x - \frac{\pi}{2} + 1 \quad (1)$$

$$\frac{dy}{dx} = \frac{v \cdot du - u \cdot dv}{\sqrt{2}}$$

Eqⁿ Normal

$$y-1 = -\frac{1}{2}(x-\pi/4)$$

$$y = -\frac{x}{2} + \frac{\pi}{8} + 1 \quad (2)$$

$$= \frac{\tan x(e^x) - e^x \sec^2 x}{\tan^2 x} \quad (2)$$

(iii) $y = \ln(\sin x)$

$$\frac{dy}{dx} = \frac{\cos x}{\sin x}$$

(d) (i) $\int \frac{x^2 - 2x}{x^3 - 3x^2 + 7} dx$

$$= \frac{1}{3} \int \frac{3x^2 - 6x}{x^3 - 3x^2 + 7} dx \left(\int \frac{f'(x)}{f(x)} \right)$$

$$= \cot x \quad (2)$$

$$= \frac{1}{3} \ln |x^3 - 3x^2 + 7| + C \quad (2)$$

(iv) $y = \ln \left[\frac{x^2 - 7x}{\sqrt{x}} \right]$

(ii) $\int_0^{\pi/8} \sec^2 2x dx$

$$y = \ln(x^2 - 7x) - \frac{1}{2} \ln x$$

$$\frac{dy}{dx} = \frac{2x-7}{x^2-7x} - \frac{1}{2x} \quad (2)$$

$$= \left[\frac{1}{2} \tan 2x \right]_0^{\pi/8}$$

(b) (i) $y = \sin^3 x$

$$= \frac{1}{2} \quad (3)$$

$$\frac{dy}{dx} = 3 \sin^2 x \cos x \quad (2)$$

(iii) $\int_1^5 (2x-1)^{1/2} dx$

$$(ii) \int 3 \sin^2 x \cos x dx = \sin^3 x \quad (2)$$

$$= \left[\frac{1}{2} \cdot \frac{2}{3} (2x-1)^{3/2} \right]_1^5$$

(c) $y = \tan x$

$$= \frac{1}{3} [27 - 1] = \frac{26}{3} \quad (3)$$

$$\frac{dy}{dx} = \sec^2 x \quad (1)$$

Q2 cont $\frac{1}{2}$ YR EXT 1 '08

$$(iv) \int_2^3 \frac{x^2}{x^3-2} dx$$

LHS = RHS $\therefore$ TRUE FOR $n=1$.

$$= \frac{1}{3} \int_2^3 \frac{3x^2}{x^3-2} dx$$

Step 2. Assume true for $n=k$

Prove true for $n=k+1$

$$= \frac{1}{3} \left[\ln |x^3-2| \right]_2^3$$

$$\text{viz } 1+8+27+\dots+k^3 = \frac{1}{4} \{k^2(k+1)^2\}$$

$$\text{R.H.S.} = \frac{1}{4} \{(k+1)^2(k+2)^2\}$$

$$\text{L.H.S.} = 1+8+27+\dots+k^3+(k+1)^3$$

$$= \frac{1}{3} \ln \frac{25}{6} \quad (3)$$

$$= \frac{1}{4} \{k^2(k+1)^2\} + (k+1)^3$$

$$= \frac{1}{4} \{(k+1)^2(k^2+4k+4)\}$$

$$= \frac{1}{4} \{(k+1)^2(k+2)^2\} = \text{R.H.S.}$$

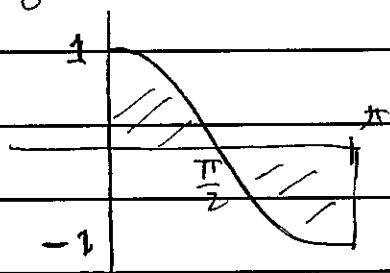
$\therefore$ True for $n=k+1$

$$(v) \int_0^\pi \cos x dx$$

Step 3. Proved true for $n=1$ Step 2

implies if true for $n=1$ true for $n=2,3,\dots$

$\therefore$ By the principle of Math Ind true for all n



Areas are equal $\therefore A=0$

(c)(i) IN $\triangle ABE$ and $\triangle ADC$

$$\angle BAE = \angle DAC \text{ (DATA)}$$

$$\angle BAE = \angle ACD (= \angle ACB) \text{ SAME ARC}$$

$$\therefore \triangle ABE \parallel \triangle ADC \text{ AAA } (2)$$

$$\text{OR } = [\sin x]_0^\pi = 0 - 0$$

$$= 0. (2)$$

$$\text{Q3 (a) } V = \pi \int_0^{\pi/4} A \tan^2 x dx$$

$$(ii) \therefore \frac{AB}{AD} = \frac{AE}{AC} \text{ RATIO SIDES SIMILAR } \triangle's$$

$$V = \pi \int_0^{\pi/4} \sec^2 x - 1 dx$$

$$\therefore AB \cdot AC = AD \cdot AE \quad (2)$$

$$= \pi [\tan x - x]_0^{\pi/4}$$

(iii) IN $\triangle ABD$ and $\triangle CED$

$$= \pi [(1 - \pi/4) - 0]$$

$$\angle BDA = \angle EDC \text{ (VERTICALLY OPP.)}$$

$$= \pi (1 - \pi/4) \text{ m}^3 \quad (3)$$

$$\angle ABC = \angle AEC \text{ (STAND SAME ARC)}$$

$$(b) 1+8+27+\dots+n^3$$

$$\therefore \triangle ABD \parallel \triangle CED \text{ (AAA) } (2)$$

$$= \frac{1}{4} \{n^2(n+1)^2\}$$

$$\text{Now } \frac{BD}{DE} = \frac{DA}{DC}$$

Step 1 Prove true for $n=1$.

$$\text{LHS} = 1 \quad \text{RHS} = \frac{1}{4} \{1^2(2^2)\} = 1$$

$$BD \cdot DC = DA \cdot DE$$

Q3 cont using result (ii)

$$AD \cdot AE + BD \cdot DC = AB \cdot AC + DA \cdot DE$$

$$AD \cdot AE - AD \cdot DE = AB \cdot AC - BD \cdot DC$$

$$AD(AE - DE) = AB \cdot AC - BD \cdot DC$$

$$AD^2 = AB \cdot AC - BD \cdot DC \quad (2)$$

$$(d) (i) \left(\frac{1}{3} + \frac{2}{3}\right)^4 = {}^4C_0 \frac{1}{3}^4 + {}^4C_1 \frac{1}{3}^3 \frac{2}{3} + {}^4C_2 \frac{1}{3}^2 \frac{2}{3}^2 + {}^4C_3 \frac{1}{3} \frac{2}{3}^3 + {}^4C_4 \frac{2}{3}^4$$

4 BLUE 3 BLUE 2 BLUE

$$t = 10 \pm \sqrt{100 - 20}$$

$$t = 5 \pm 2\sqrt{5}$$

$$\tan \theta/2 = 5 \pm 2\sqrt{5}, \theta/2 = 1.4656, 0.4856$$

$$\theta = 0.97^\circ, 2.93^\circ \quad (4)$$

$$(d) (i) y = 2^x$$

$$y = e^{(\ln 2)x}$$

$$\frac{dy}{dx} = \ln 2 \cdot e^{\ln 2 x}$$

$$= \ln 2 (2^x) \quad (2)$$

$$(ii) \int_1^3 2^x dx$$

$$= \left[\frac{2^x}{\ln 2} \right]_1^3$$

$$= \frac{6}{\ln 2}$$

$$(e) (i) \# = \frac{9!}{2!2!} = 90720 \quad (2)$$

$$(ii) 2R's + 2E's + 1 \text{ other letter} = 1 \times 1 \times 5 = 5 \text{ Selections}$$

$$2R's \text{ or } 2E's + 3 \text{ other letters} = 2 \times 6 = 40$$

$$\text{no doubles} \quad {}^7C_5 = 210$$

$$\therefore 66 \text{ SELECTIONS} \quad (2)$$

Arrangements

$$\textcircled{1} \frac{5!}{2!2!} = 30, 30 \times 5 = 150$$

$$\textcircled{2} \frac{5!}{2!} = 60, 60 \times 40 = 2400$$

$$\textcircled{3} \frac{5!}{5!} = 120, 120 \times 21 = 2520$$

$$\text{TOTAL ARRANGEMENTS} = 5070 \quad (2)$$

Question 4.

(a) (i) IN QUAD ACEM.

$$\angle CAM = 90^\circ \text{ (DATA)}$$

$$\angle MEN = 90^\circ \text{ (STAND ON DIAMETRE)}$$

$\therefore$ ACEM IS A CYCLIC QUAD

$$(\text{EXT } \angle = \text{INT OPP } \angle)$$

(ii) IN $\triangle CAN$ AND $\triangle MEN$

$$\angle CAN = \angle MEN \quad (90^\circ)$$

$$\angle CNA \text{ IS COMMON}$$

$$\therefore \angle ACN = \angle EMN \quad (3^{\text{RD}} \text{ OF } \triangle)$$

$$\text{BUT } \angle NDE = \angle EMN \quad (\text{SAME ARC})$$

$$\therefore \angle ACN = \angle NDE$$

$\therefore$ QUAD BCED IS CONCYCLIC

$$(\text{OPP INT } \angle = \text{EXT } \angle) \quad (3)$$

$$(b) \left(2x^2 - \frac{3}{x}\right)^9 = \dots {}^9C_6 8x^6 \cdot \left(\frac{-3}{x}\right)^3$$

$$\text{TERM} = {}^9C_6 8 \times 729$$

$$= 489888$$

Question 5

$$(a) f'(x) = \lim_{h \rightarrow 0} \frac{\frac{1}{x+h} - \frac{1}{x}}{h}$$

$$= \lim_{h \rightarrow 0} \frac{x - (x+h)}{h(x+h)(x)}$$

$$= \lim_{h \rightarrow 0} \frac{-h}{h(x+h)(x)}$$

$$= \lim_{h \rightarrow 0} \frac{-1}{(x+h)(x)}$$

$$= \frac{-1}{x^2} \quad (3)$$

$$(b) (i) f(x) = \sec x$$

$$= \frac{1}{\cos x}$$

$$f(x) = (\cos x)^{-1}$$

$$f'(x) = -1(\cos x)^{-2} \cdot -\sin x$$

$$= \frac{\sin x}{\cos^2 x}$$

$$= \frac{\tan x}{\cos x}$$

$$= \tan x \sec x \quad (2)$$

$$(ii) f''(x) = v \frac{du}{dx} + u \cdot \frac{dv}{dx}$$

$$= \sec x \sec^2 x + \tan^2 x \sec x$$

$$= \sec x (\sec^2 x + \tan^2 x) \quad (2)$$

$$(iii) f''(x) = \sec x (2 \tan^2 x + 1)$$

$$\text{for } \text{cup } f''(x) > 0$$

$$f''(x) > 0 \quad \sec x > 0$$

$$0 < \theta \leq \frac{\pi}{2} \quad \frac{3\pi}{2} < \theta \leq 2\pi$$

$$(c) 3 \tan^5 \theta - 10 \tan^3 \theta + 3 \tan \theta = 0$$

$$\tan \theta (3 \tan^4 \theta - 10 \tan^2 \theta + 3) = 0$$

$$\therefore \tan \theta = 0 \quad \theta = 0, 180, 360$$

$$\theta = 0, \pi, 2\pi$$

$$\text{OR } 3 \tan^4 \theta - 10 \tan^2 \theta + 3 = 0$$

$$(3 \tan^2 \theta - 1)(\tan^2 \theta - 3) = 0$$

$$\tan^2 \theta = \frac{1}{3} \quad \tan^2 \theta = 3$$

$$\tan \theta = \pm \frac{1}{\sqrt{3}} \quad \tan \theta = \pm \sqrt{3}$$

$$\theta = \frac{\pi}{6}, \frac{\pi}{3}, \frac{2\pi}{3}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{4\pi}{3}, \frac{5\pi}{3}, \frac{11\pi}{6}$$

$$+ 0, \pi, 2\pi$$

$$(4)$$

$$(d) (i) \frac{1}{x-2} - \frac{1}{x+3} = \frac{x+3 - (x-2)}{(x-2)(x+3)}$$

$$= \frac{5}{(x-2)(x+3)} \quad (1)$$

$$(ii) \int_3^5 \frac{1}{(x-2)(x+3)} dx$$

$$= \frac{1}{5} \int_3^5 \frac{5}{(x-2)(x+3)} dx$$

$$= \frac{1}{5} \int_3^5 \left(\frac{1}{x-2} - \frac{1}{x+3} \right) dx$$

$$= \frac{1}{5} \left[\ln |x-2| - \ln |x+3| \right]_3^5$$

Question 5 cont.

$$= \frac{1}{5} \{ (\ln 3 - \ln 8) - (\ln 1 - \ln 6) \}$$

$$= \frac{1}{5} \ln \frac{3 \times 6}{8}$$

$$= \frac{1}{5} \ln \frac{18}{8} = \frac{1}{5} \ln \frac{9}{4}$$

$$= \frac{1}{5} \ln \left(\frac{3}{2}\right)^2$$

$$\textcircled{3} = \frac{2}{5} \ln \frac{3}{2} u^2$$

(e) (i) $2 \sin \theta + 2\sqrt{3} \cos \theta$

$$= R \sin(\theta + \alpha)$$

$$\therefore R \sin \theta \cos \alpha + R \cos \theta \sin \alpha$$

$$= 2 \sin \theta + 2\sqrt{3} \cos \theta$$

$$R \cos \alpha = 2 \quad R \sin \alpha = 2\sqrt{3}$$

$$\frac{R \sin \alpha}{R \cos \alpha} = \tan \alpha = \sqrt{3}$$

$$\alpha = \pi/3$$

and $R^2 \cos^2 \alpha = 4$

$$R^2 \sin^2 \alpha = 12$$

$$R^2 (\cos^2 \alpha + \sin^2 \alpha) = 16$$

$$R^2 = 16 \quad R = 4$$

$$2 \sin \theta + 2\sqrt{3} \cos \theta$$

$$= 4 \sin(\theta + \pi/3)$$

$\textcircled{3}$

5 millimetre Grid

